

Name: _____

Class: _____

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12

HSC ASSESSMENT TASK 3

JUNE 2012

MATHEMATICS Extension 1

Time Allowed: 70 minutes

Instructions:

- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new page**.
- Standard integrals can be found on the last page.
- Write in blue or black pen only

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/10	/10	/10	/10	/10	/10	/60

Question 1	(10 marks)	Marks
a)	Find the exact value of $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$	1
b)	Differentiate the following	
(i)	$y = x e^{x^2}$	2
(ii)	$y = \cos^{-1} 3x$	2
(iii)	$y = \log_e \frac{x+4}{x^2}$	2
(iv)	$y = \log_a x$	1
c)	Simplify $\log_a x^4 \div \log_a x^{\frac{2}{3}}$	2

Question 2	(10 marks)	Marks
Consider the function $f(x) = \frac{x}{x-3}$		
(i)	Show that $f'(x) < 0$ for all x in the domain	2
(ii)	Find the equation of the horizontal asymptote	1
(iii)	Without using any further calculus sketch the graph of $y = f(x)$ showing asymptotes and the x and y intercepts	2
(iv)	Explain why $f(x)$ has an inverse function $f^{-1}(x)$	1
(v)	Find an expression for $f^{-1}(x)$	2
(vi)	For what value(s) of x does $f(x) = f^{-1}(x)$?	2

Question 3	(10 marks)	Marks
a)	Sand is poured into a conical heap at a constant rate of $0.6 \text{ m}^3/\text{s}$ so that the height of the heap is always equal to twice the radius of the base.	
(i)	Show that $\frac{dV}{dr} = 2\pi r^2$	1
(ii)	When the heap is 5m high, find the rate of increase of the radius of the base.	2
b)	Find	
(i)	$\int \frac{x}{x^2+4} dx$	1
(ii)	$\int \frac{1}{x^2+4} dx$	2
(iii)	$\int \frac{dx}{\sqrt{1-4x^2}}$	2
c)	Sketch $y = \sin^{-1}(2x - 1)$ showing all important features	2

Question 4	(10 marks)	Marks
a)	Find m if $x^{m+2} = e^{6\ln x}$ where $x > 0$	2
b)	Find the general solution of $2 \sin x = -1$	2
c)	Consider the function $y = \frac{\ln x}{x}$ where $x > 0$	
(i)	The function has a maximum turning point in the domain $x > 0$. Find its coordinates.	2
(ii)	Find $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$	1
(iii)	Sketch the curve showing all important features	2
d)	Let $\tan^{-1} x = \alpha$. Find an expression for $\cos \alpha$ in terms of x given that α is acute.	1

Question 5**(10 marks)****Marks**

- a) Evaluate $\int_1^3 \frac{dx}{(1+x)\sqrt{x}}$ using the substitution $u = \sqrt{x}$ or otherwise. 3
Leave your answer in exact form.
- b) Find $\int \cos^2 \frac{x}{2} dx$ 2
- c) Given $f(x) = 3 \cos^{-1}(\sin 2x) - 2 \sin^{-1}(\cos 3x)$, show that $f(x)$ is a constant function by finding $f'(x)$ 3
- d) Using the table of standard integrals, find the exact value of 2

$$\int_0^{\frac{\pi}{8}} \sec 2x \tan 2x dx$$

Question 6**(10 marks)****Marks**

- a)
- (i) Express $\sin x - \cos x$ in the form $R \sin(x - \alpha)$ where $R > 0$ and α is acute. 2
- (ii) Hence solve $\sin x - \cos x = -1$ over the domain $0 \leq x \leq 2\pi$ 2
- b) Assume that at time t years the population $P(t)$ of a town is given by $P(t) = 50000 + Ae^{kt}$ where A and k are constants.
- (i) Show that $P(t)$ satisfies the equation $P'(t) = k [P(t) - 50000]$ 1
- (ii) Given that $P(0) = 70000$, evaluate A 1
- (iii) If $P(8) = 150000$, find k correct to 4 decimal places 2
- (iv) How many years will it take for the population to reach 200000? (give your answer correct to the nearest integer) 2

2012 Ext. 1 June Ass. Task Solutions

Question 1

$$\begin{aligned} \text{a) } \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) &= x \\ \cos x &= -\frac{\sqrt{3}}{2} \\ \therefore x &= \frac{5\pi}{6} \end{aligned}$$

$$\begin{aligned} \text{b) i) } y &= x e^{x^2} \\ y' &= x \times 2x e^{x^2} + e^{x^2} \\ &= 2x^2 e^{x^2} + e^{x^2} \end{aligned}$$

$$\begin{aligned} \text{c) i) } y &= \cos^{-1} 3x \\ y' &= \frac{-1}{\sqrt{1-(3x)^2}} \times 3 \\ &= \frac{-3}{\sqrt{1-9x^2}} \end{aligned}$$

$$\begin{aligned} \text{c) ii) } y &= \log_e \frac{x+4}{x^2} \\ &= \log_e(x+4) - \log_e x^2 \\ y' &= \frac{1}{x+4} - \frac{2x}{x^2} \\ &= \frac{1}{x+4} - \frac{2}{x} \end{aligned}$$

$$\begin{aligned} \text{c) iv) } y &= \log_a x \\ &= \frac{\log_e x}{\log_e a} \\ y' &= \frac{1}{\log_e a} \times \frac{1}{x} \\ y' &= \frac{1}{x \log_e a} \end{aligned}$$

$$\begin{aligned} \text{c) } \log_a x^4 &\div \log_a x^{\frac{2}{3}} \\ 4 \log_a x &\div \frac{2}{3} \log_a x \\ &= \frac{4}{\frac{2}{3}} \\ &= 4 \times \frac{3}{2} = \underline{6} \end{aligned}$$

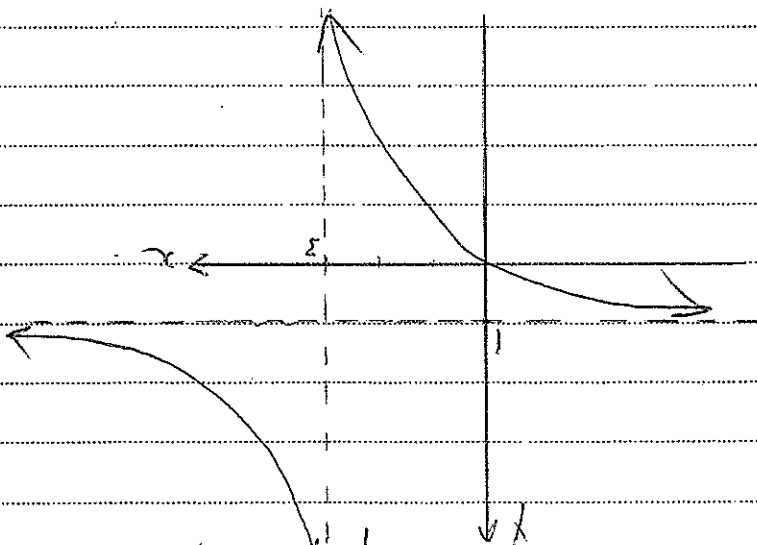
Question 2

$$\begin{aligned} \text{a) i) } f(x) &= \frac{x}{x-3} \\ f'(x) &= \frac{(x-3) \times 1 - x \times 1}{(x-3)^2} \\ f'(x) &= \frac{-3}{(x-3)^2} < 0 \\ &\text{for all } x \end{aligned}$$

$$\begin{aligned} \text{c) ii) } \text{As } x &\rightarrow \infty \\ f(x) &\rightarrow 1 \uparrow \\ \therefore y &= 1 \text{ is horizontal asymptote} \end{aligned}$$

(iv) Satisfies horizontal line test.

(iii) Vertical asymptote $x = 3$. Intercept $(0,0)$



$$(v) \quad y = \frac{x-3}{x^2-3x}$$

$$x = \frac{y+3}{y}$$

$$xy - 3x = y$$

$$xy - y = 3x$$

$$y(x-1) = 3x$$

$$y = f^{-1}(x) = \frac{3x}{x-1}$$

(vi) $f(x) = f^{-1}(x)$ on $y = x$

$$\frac{x}{x-3} = x$$

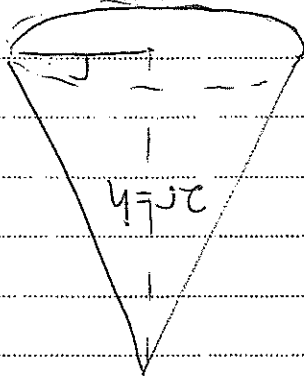
$$x = x^2 - 3x$$

$$0 = x^2 - 4x$$

$$0 = x(x-4)$$

$$A+ \quad x=0 \text{ or } 4$$

Question 3



$$\frac{dV}{dt} = 0.6$$

$$(i) \quad V = \frac{1}{3} \pi r^2 h$$

$$= \frac{2}{3} \pi r^3$$

$$\frac{dV}{dt} = 2\pi r^2$$

$$(ii) \quad \frac{dV}{dt} \times \frac{dt}{dr} = \frac{dV}{dr}$$

$$\frac{dV}{dt} \times \frac{dt}{dr} = \frac{dV}{dr}$$

$$\frac{dV}{dt} \times \frac{dt}{dr} = \frac{dV}{dr}$$

$$\frac{1}{2\pi} \times 9.0 =$$

$$\frac{dr}{dt} = 0.015 \text{ m/s}$$

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$$b) \text{ (i) } \int \frac{x}{x^2+4} dx$$

$$= \underline{\underline{\frac{1}{2} \log(x^2+4) + C}}$$

$$\text{c) } \int \frac{1}{x^2+4} dx$$

$$= \frac{1}{2} \int \frac{2}{x^2+2^2} dx$$

$$= \underline{\underline{\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C}}$$

$$\text{c) } \int \frac{dx}{\sqrt{1-4x^2}}$$

$$\int \frac{dx}{\sqrt{\left(\frac{1}{2}\right)^2 - x^2}} \cdot 4$$

$$\frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{1}{2}\right)^2 - x^2}}$$

$$= \frac{1}{2} \sin^{-1}\left(\frac{x}{\frac{1}{2}}\right)$$

$$= \underline{\underline{\frac{1}{2} \sin^{-1}(2x) + C}}$$

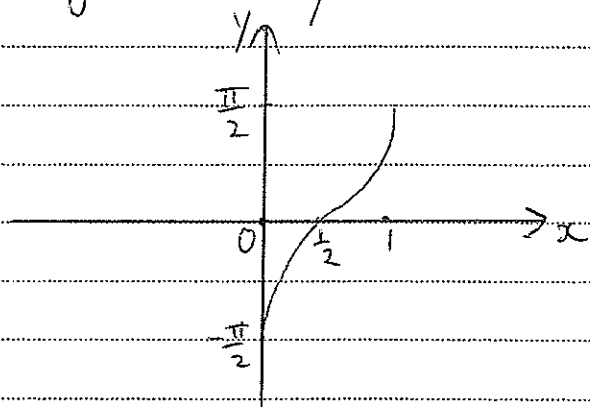
$$\text{c) } y = \sin^{-1}(2x-1)$$

$$\text{Domain: } -1 \leq 2x-1 \leq 1$$

$$0 \leq 2x \leq 2$$

$$0 \leq x \leq 1$$

$$\text{Range: } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



Question 4

$$\text{a) } x^{m+2} = e^{6 \ln x}$$

$$x^{m+2} = e^{\ln x^6}$$

$$x^{m+2} = x^6$$

$$\underline{\underline{m = 4}}$$

$$\text{b) } 2 \sin x = -1$$

$$\sin x = -\frac{1}{2}$$

$$\sin x = \sin\left(-\frac{\pi}{6}\right)$$

$$x = n\pi + (-1)^n \left(-\frac{\pi}{6}\right)$$

where n is an integer

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$$c) i) y = \frac{\ln x}{x}$$

$$\frac{dy}{dx} = \frac{x \times \frac{1}{x} - \ln x}{x^2}$$

$$0 = \frac{1 - \ln x}{x^2} \text{ for stat. pt.}$$

$$\therefore 1 - \ln x = 0$$

$$\ln x = 1$$

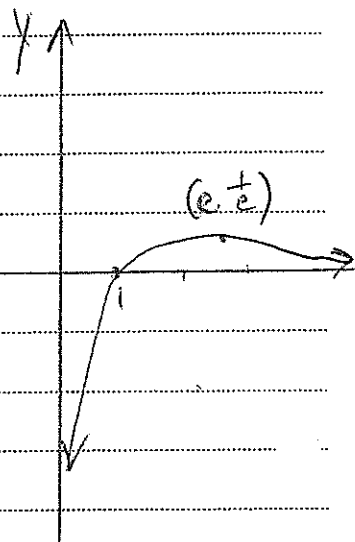
$$e^1 = x$$

$$x = e$$

$(e, \frac{1}{e})$ is stat. pt.

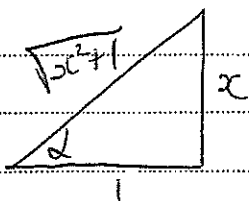
$$ii) \lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$$

(iii)



$$d) \tan^{-1} x = \alpha$$

$$\therefore x = \tan \alpha$$



$$\cos \alpha = \frac{1}{\sqrt{x^2 + 1}}$$

Question 5

$$a) \int_1^3 \frac{dx}{(1+x)\sqrt{x}}$$

$$\text{Let } u = \sqrt{x}$$

$$du = \frac{dx}{2\sqrt{x}}$$

$$\int_1^{\sqrt{3}} \frac{2u du}{(1+u^2)u}$$

$$2 [\tan^{-1} u]_1^{\sqrt{3}}$$

$$= \frac{\pi}{6}$$

$$b) \int \cos^2 \frac{x}{2} dx$$

$$\int \frac{\cos x + 1}{2} dx$$

$$\frac{1}{2} [\sin x + x]$$

$$\frac{\sin x}{2} + \frac{x}{2} + C$$

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$$c) f(x) = 3\cos^{-1}(\sin 2x) - 2\sin^{-1}(\cos 3x)$$

$$f'(x) = \frac{-3 \times 2 \cos 2x}{\sqrt{1 - \sin^2 2x}} - \frac{2 \times -3 \sin 3x}{\sqrt{1 - \cos^2 3x}}$$

$$= \frac{-6 \cancel{\cos 2x}}{\cancel{\cos 2x}} + \frac{6 \cancel{\sin 3x}}{\cancel{\sin 3x}}$$

$$= 0 \quad \therefore f(x) \text{ must be a constant}$$

$$d) \int_0^{\frac{\pi}{8}} \sec 2x + \tan 2x \, dx$$

$$= \left[\frac{1}{2} \sec 2x \right]_0^{\frac{\pi}{8}}$$

$$= \frac{1}{2} \sec \frac{\pi}{4} - \frac{1}{2} \sec 0$$

$$= \frac{1}{2 \cos \frac{\pi}{4}} - \frac{1}{2 \cos 0}$$

$$= \frac{1}{\frac{2}{\sqrt{2}}} - \frac{1}{2}$$

$$= \frac{\sqrt{2}}{2} - \frac{1}{2}$$

$$= \frac{\sqrt{2} - 1}{2}$$

Question 6

$$a) \text{ ci) } \sin x - \cos x \equiv A \sin(x - \alpha)$$

$$A = \sqrt{1^2 + 1^2}, \quad \alpha = \tan^{-1} \frac{1}{1}$$

$$= \frac{\pi}{4}$$

$$\therefore \sin x - \cos x = \sqrt{2} \sin\left(x - \frac{\pi}{4}\right).$$

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$$\text{cii)} \quad \sqrt{2} \sin\left(x - \frac{\pi}{4}\right) = -1$$

$$\sin\left(x - \frac{\pi}{4}\right) = \frac{-1}{\sqrt{2}}$$

$$x - \frac{\pi}{4} = \frac{5\pi}{4}, \frac{7\pi}{4} \text{ also } -\frac{\pi}{4}$$

$$\therefore \underline{x = 0, \frac{3\pi}{2}, 2\pi}$$

$$\text{c) ci)} \quad P(t) = 50000 + Ae^{kt}$$

$$P'(t) = kAe^{kt}$$

$$= k(P(t) - 50000)$$

$$\text{cii)} \quad 70000 = 50000 + Ae^{kt}$$

$$20000 = Ae^{kt}$$

$$\underline{20000 = A \text{ since } e^0 = 1}$$

$$\text{ciii)} \quad 150000 = 50000 + 20000e^{8t}$$

$$100000 = 20000e^{8t}$$

$$5 = e^{8k}$$

$$\log_e 5 = 8k$$

$$k = \log_e 5 \div 8$$

$$\underline{k = 0.2012 \text{ to 4 d.p.}}$$

$$\text{civ)} \quad 200000 = 50000 + 20000e^{0.2012t}$$

$$150000 = 20000e^{0.2012t}$$

$$7.5 = e^{0.2012t}$$

$$\log_e 7.5 = 0.2012t$$

$$\underline{t = 10 \text{ years}}$$